

Worksheet 1 *Tabulating change*

Prepares you for lab manual pages **BT1, BT2, Pop1, Pop2, SL1, SL2** (section L1 S2)

Name:

Name of TA:

This worksheet is about using *tables* to track a changing quantity. Our basic table has columns for a quantity (X), time (t), the time step (Δt), the change in the quantity (ΔX), and the rate of change $\Delta X/\Delta t$. The Greek letter Δ means “change”: $\Delta X = [\text{next value of } X] - [\text{current value of } X]$. The change columns are blank in the last row, because we do not know what happens next. Example: the height X of a tree, observed every six months.

X (m)	t	ΔX (m)	Δt (months)	$\Delta X/\Delta t$ (m per month)
50	Jan 2020	5	6	0.833
55	Jul 2020	3	6	0.5
58	Jan 2021	2	6	0.33
60	Jul 2021			

The growth of X is *slowing*, since the rate of change decreases (0.833, 0.5, 0.33). **In each table below, fill in every blank cell that the given information allows you to compute.** Some cells are given; the change columns of the last row stay blank, since no later observation exists.

1. F is the number of flowers in a garden, counted on several observation days; t is time.

F	t	ΔF	Δt (days)	$\Delta F/\Delta t$ (flowers/day)
120	March 1			
130	March 2			
170	March 4			
175	March 5			

2. C is the thickness of an iceberg in meters; t is time. Here the *changes* were recorded, and you must work out the thickness.

C	t	ΔC	Δt (days)	$\Delta C/\Delta t$ (m/day)
35	Dec. 1	−0.5		
	Dec. 15	−0.3		
	Dec. 19	0.2		
	Jan. 2			

3. M is the amount (mg) of an injected drug in the bloodstream; t is hours since injection. Assume $\Delta M/\Delta t = -0.1 M$ in every row, and use it to work forward one hour at a time.

M (mg)	t	ΔM	Δt (hours)	$\Delta M/\Delta t$ (mg/hr)
80	0			−8
	1			
	2			
	3			